CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 4: Artificial Neural Networks & Backpropagation

Course: Program Elective Course-I [23CS3507]
Category: Assignments | Assignment #4 | Academic Year: 2026-27

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1. Overview & Objective:

Perceptron models, multi-layer networks, activation functions, and gradient backpropagation.

2. Assignment Problems & Instructions:

1. Prove why a single-layer perceptron cannot classify the non-linear XOR function.
2. Derive the backpropagation weight update formula using multivariate chain rule.
3. Compare activation functions: Sigmoid, Hyperbolic Tangent Tanh , and Rectified Linear Unit ReLU .
4. Explain the vanishing gradient problem in deep networks and techniques to mitigate it.

Submission Guidelines & Verification:

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